COEUS

Semantic Web Application Framework

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http://bioinformatics.ua.pt

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INSTITUTO DE ENGENHARIA ELECTRÓNICA
E TELEMÁTICA DE AVEIRO



UNIVERSIDADE DE AVEIRO



- *What?*
- *Why?*
- *How?*
- *Setting it up...*
- *Where?*
- *Closing*

What?

Next Generation Semantic Web Application Framework

Rapid Application Deployment

Data Integration Platform

Advanced Data Exploration Framework

"SEMANTIC WEB IN A BOX" FOR BIOINFORMATICS

Why?

(Among others...)

IMPROVE BIOMEDICAL APPLICATIONS DOMAIN

SEMANTIC WEB + LIFE SCIENCES

perfect benchmark

awareness from bioinformatics developers

common intertwined graph **relationship structure**

BUT...

Semantic Web is still underused *in the life sciences domain*

transition/migration is very difficult

complex application deployment

learn a whole new batch of technologies (again!)

You *are notable exceptions!*

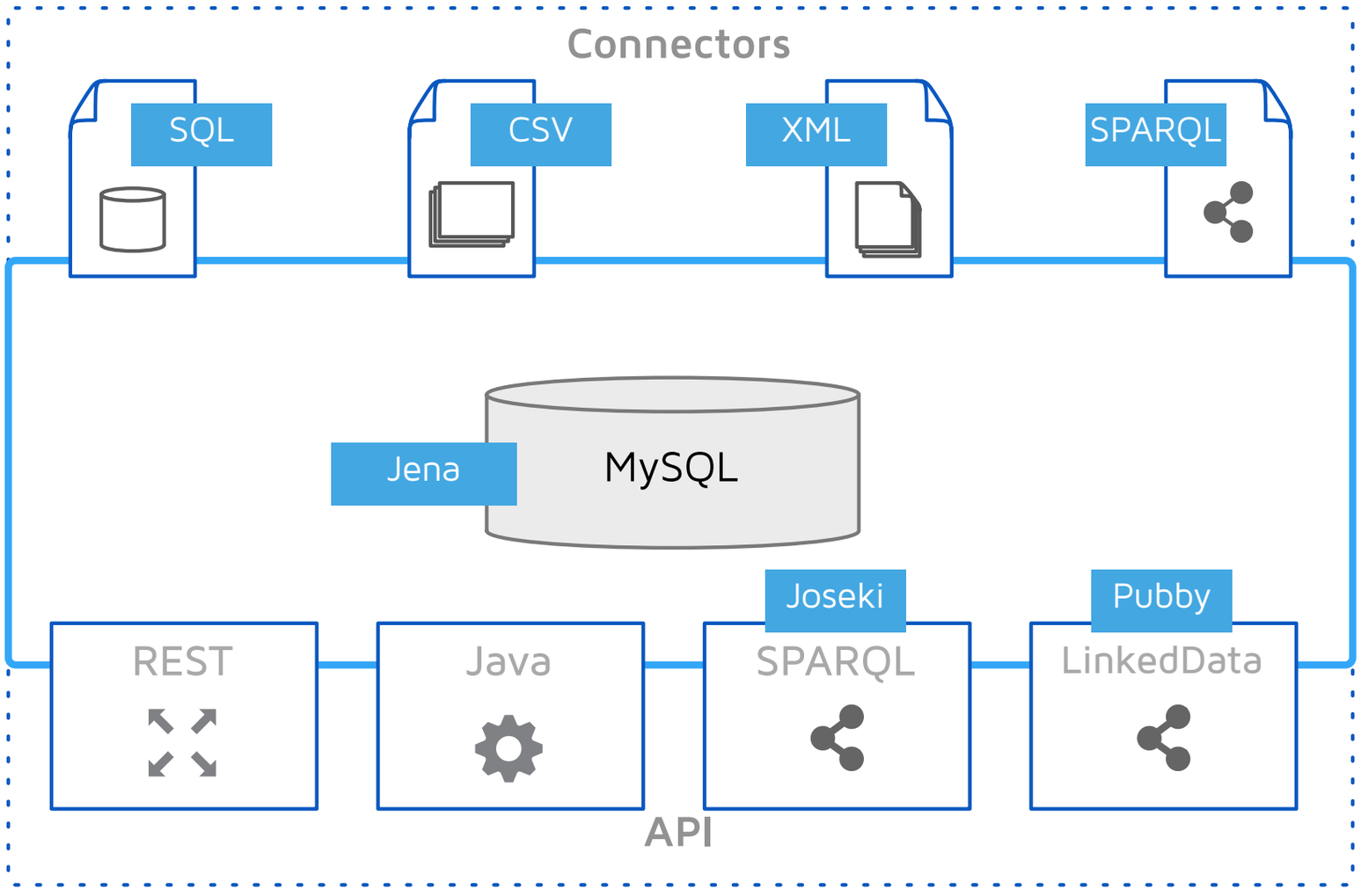
How?



KNOWLEDGE

SEED

is a single app, an instance

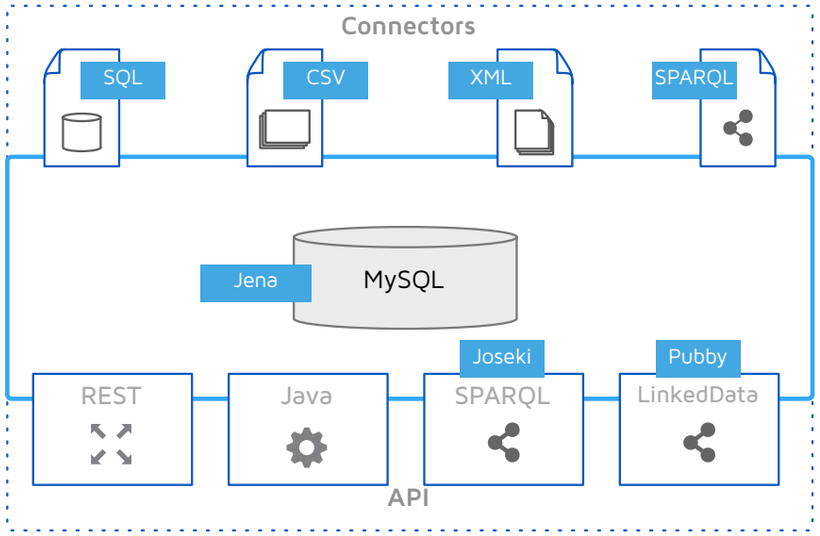


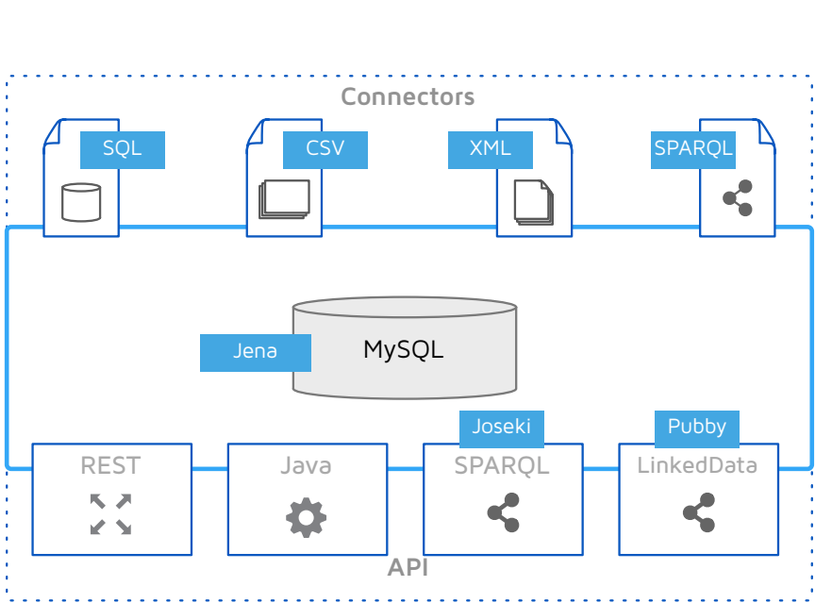
KNOWLEDGE
SEED

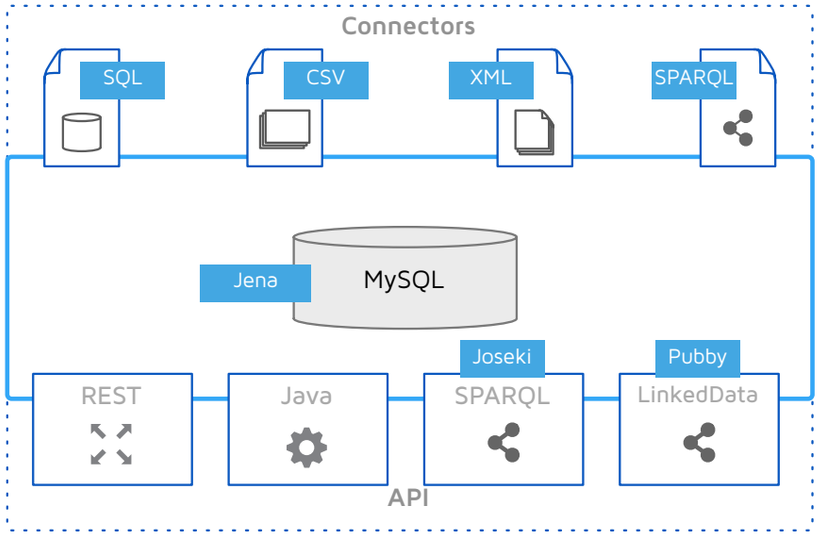
is a single app, an instance

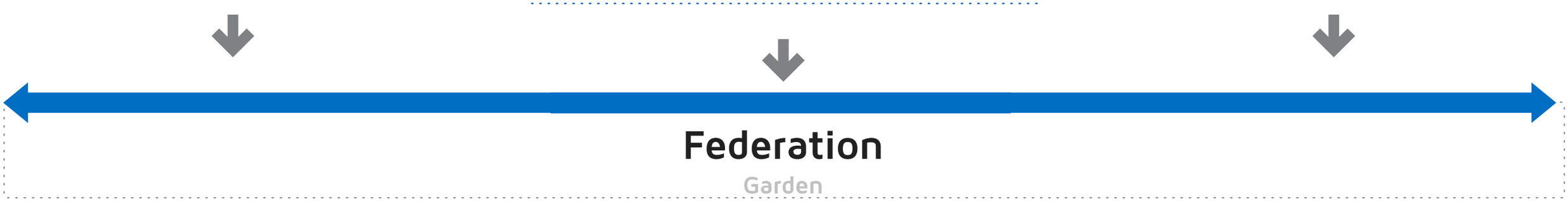
KNOWLEDGE
GARDEN

links multiple seeds











INTEGRATION

Data in = Connectors

CONNECTORS = CSV + XML + SQL + SPARQL

load everything

flexible format support

map external data **into** a seed's model

any predicate matches data from *selectors*

csv **columns**

xml **XPath** results

SQL **table** columns

SPARQL query **variables**

Data out = APIs

EXPLORATION

API = SPARQL + REST + Java + LinkedData

code your own apps

in **any** programming language available

for desktop, **web** or **mobile**

one seed, multiple apps and/or *multiple seeds, one app*

federate to existing apps

access with **REST** or **SPARQL**

distributed knowledge network

Setting it up...



http://bioinformatics.ua.pt/COEUS

CONFIGURATION

Where's everybody?

CONFIGURATION = JS FILE + (ONTOLOGY * SETUP)

local app configuration

JSON object in JavaScript file

defines application **properties** (name, version, environment…)

define seed ontology then *setup resources*

seed ontology

based on COEUS ontology

your app **data model**

resource setup

resource **connectors** configuration

ontology **mappings**

Getting data together...

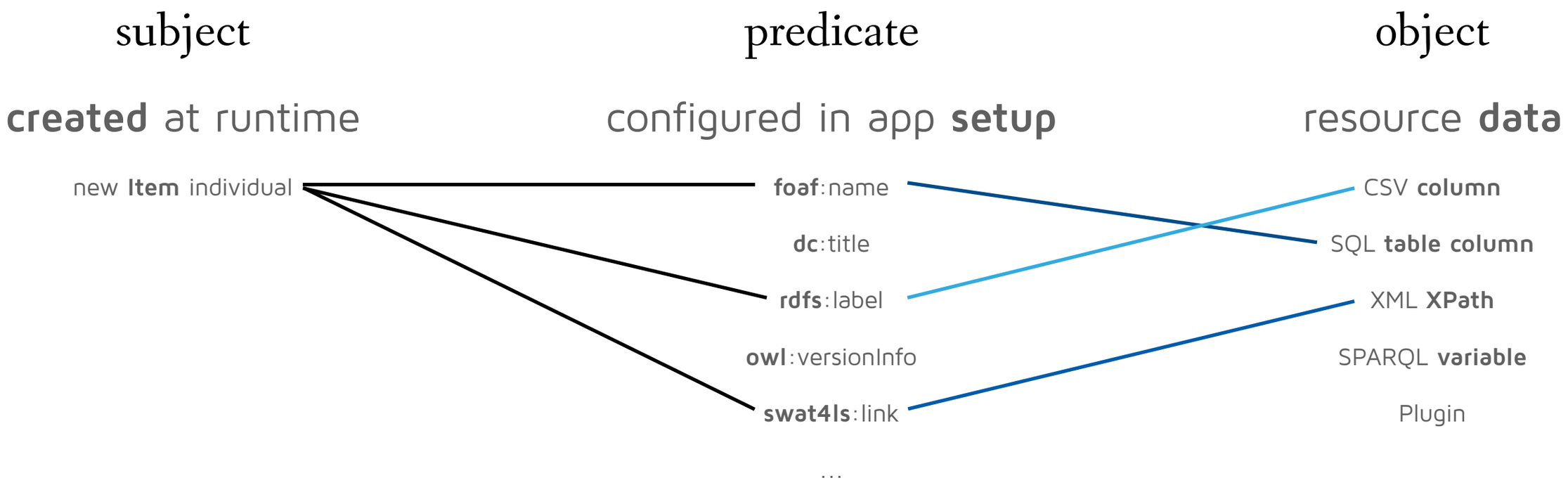
BUILDING

SEMANTIC WEB TRANSLATION

triplify data

app setup + data = **triples**

subject - predicate - *object*



Putting the Web *in Semantic Web*

EXPLORING

INTEROPERABILITY

SPARQL

…/`sparql`

perform SPARQL **queries**

get data in multiple **formats**

GET requests for all data

REST

…/`api/<concept>_<id>` for **relationships**

…/`api/sub/pred/obj/<format>` for specific **content**

Java

abstracted from **Jena**'s methods

LinkedData

…/`resource/<concept>_<id>`

symfony

php

Microsoft Visual Studio

Sencha

PhoneGap

android

django

jQuery
write less, do more.

BioPerl

Java

iOS 5

Windows Phone
Put people first.

RAILS

zepto.JS

catalyst
Web Framework

Where?

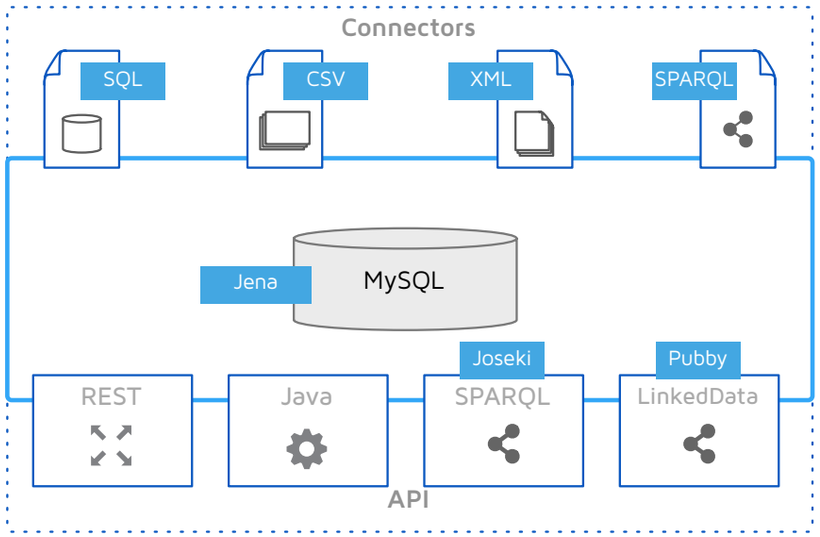


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Drugs

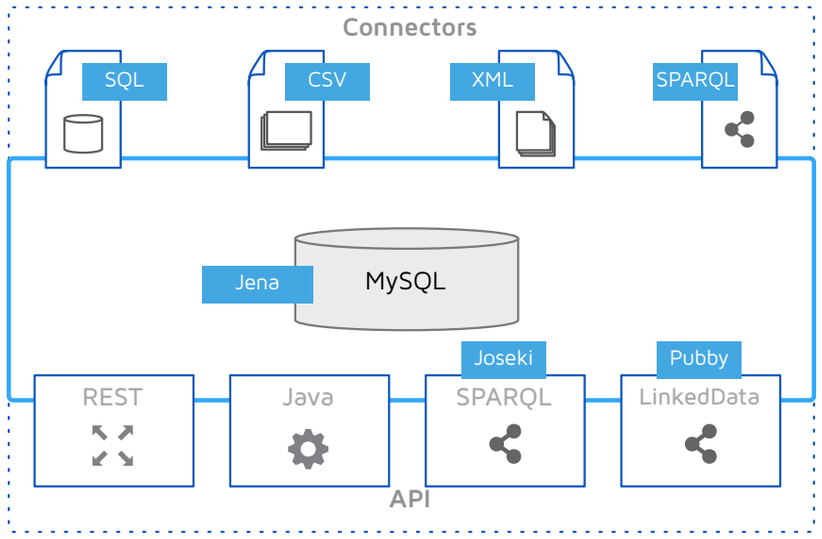
App

Here!



Diseasecard

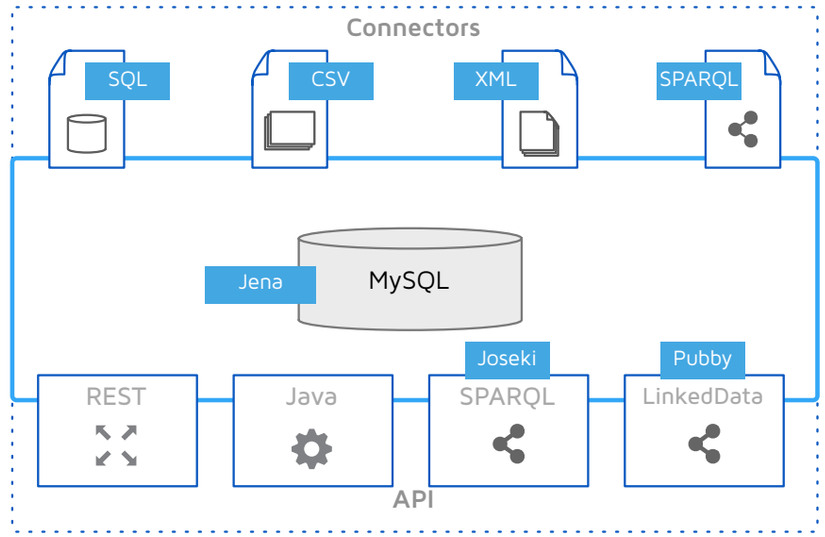
http://bioinformatics.ua.pt/**dc4**

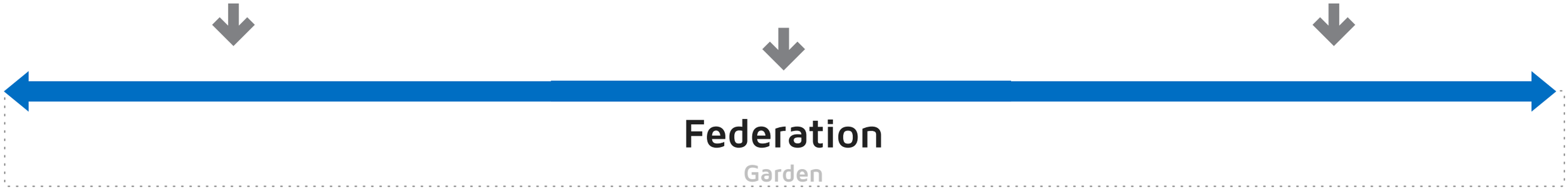


Whatever

App

Here!





Knowledge Here!

Knowledge Here!

Knowledge Here!

Knowledge Here!

Knowledge Here!

Knowledge Here!


Closing

INTEGRATION

Connectors CSV, SQL, XML, SPARQL

+

EXPLORATION

API Java, REST, SPARQL, LinkedData

=

"SEMANTIC WEB IN A BOX"

COEUS
ENABLING KNOWLEDGE

http://bioinformatics.ua.pt/**coeus**

Fork me on GitHub

WE ARE LOOKING FOR RESEARCH COLLABORATIONS!

THANK YOU!

Questions and Feedback are (very) *welcome!*

http://bioinformatics.ua.pt/COEUS